

Name:	
Class:	
Date:	

Honor pledge <i>“On my honor, I have neither given nor received unauthorized aid.”</i> Student Signature: _____
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Subject:	IGCSE Computer Science
Exam:	G10 T1 A1 (Unit 6 Software) QP
Exam Duration:	45 Minutes
Teacher:	Mr Ellis

Instructions to Students:
Total marks = 40

Cambridge Secondary School Grade Conversion Matrix		
Cambridge Grade	ISR % Grade	Descriptive
A*	90-100	Outstanding
A	85-89	Excellent
B	80-84	Good
C	70-79	Satisfactory
D	60-69	Needs Improvement
E	45-59	Unsatisfactory
F	0-44	Unsatisfactory

Marks:
ISR % Grade:
Cambridge Grade:

Assessment test

Unit 6 Software

1. One function of a modern PCs operating system is multi-tasking. This allows two or more programs to be open at the same time.

(a) Describe how multi-tasking works.

[4]

(b) When a user presses a key on their keyboard an interrupt is generated. Describe the actions of the operating system as a result of this interrupt.

[2]

(c) A keypress on a keyboard is an example of a hardware interrupt.

Give **one** example of a software interrupt.

[1]

- (d) Other than managing multi-tasking and interrupts, explain **three** other functions of an operating system

[6]

2. A disk defragmenter is an example of a utility program.

(a) Explain what a utility program is.

[2]

(b) A disk defragmenter may improve the performance of a computer.

Describe how a disk defragmenter works to improve the performance of a computer. [3]

(c) Hard drives normally use magnetic or solid-state technologies.

Explain which technology will need defragmentation.

[2]

Assessment test

Unit 6 Software

3. Zach buys a new computer with an operating system installed. When he first turns it on, the computer runs a bootloader.

(a) The bootloader is firmware. Explain what the term firmware means. [2]

(b) State the purpose of a bootloader. [1]

Assessment test

Unit 6 Software

4. A programmer is creating a device driver for a new webcam.

They make use of assembly language to produce the device driver.

- (a) Identify whether assembly language is a high-level or low-level language. [2]

- (b) The assembly code produced by the programmer makes use of mnemonics.

Give the purpose of using mnemonics when creating assembly code. [1]

- (c) Give **one** reason why the programmer has chosen assembly language rather than a high-level language to write the device driver. [1]

- (d) Once the programmer has completed their assembly program, state the translator they will need to use to create the machine code. [1]

The programmer is also making an application that will allow users to edit videos made with their webcam.

- (e) Explain why a high-level language would be more appropriate than assembly code for programming this application. [2]

Assessment test

Unit 6 Software

- (f) One the video editing application is complete the programmer makes use of a compiler. Describe the purpose of a compiler in this situation. [2]

- (g) The programmer could have made use of a high-level language that would require an interpreter rather than a compiler. [1]
State the meaning of the term interpreter.

- (h) Explain why an interpreter wouldn't be as good a choice as a compiler for the video editing application. [2]

5. Most programmers make use of IDE when writing program code.

(a) One key feature of an IDE is a code editor.

Describe **two** features that are available to a programmer that will change how the code appears in the code editor. [4]

(b) Name **one** other feature of an IDE that will help the programmer with debugging their code. [1]

[Total 40 marks]